

CourierXpress

Supervisor: *LÊ THANH NHÂN*

Semester: *I*

Batch No: *T52311.M0*

Group No: *06*

Order:	Full name	Roll No.
<i>1.</i>	<i>Vuong Que Han</i>	<i>Student1665675</i>
<i>2.</i>	<i>Le Hoang Thong</i>	<i>Student1645283</i>
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Month:4 Year: 2026

This is to certify that

Mr. **Phan Van Duy**

Mr. **Le Hoang Thong**

Ms. **Vuong Que Han**

Have successfully Designed & Developed

CourierXpress

Submitted by:

Mr. LE THANH NHAN

Date Of Issue:

Authorized Signature:

Content

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Acknowledge

As we understand that the Project is a step-by- step learning environment that closely simulates the class-room and Lab based learning environment into actual implementation. It is a project implementation at your fingertips!! An electronic, live juncture on the machine that allows you to.

- ✓ Practice step by step i.e. ladder approach.
- ✓ Build a larger more robust application.
- ✓ Usage of certain utilities in applications designed by user.
- ✓ Single program to unified code leading to a complete application.
- ✓ Learn implementation of concepts in a phased manner.
- ✓ Enhance skills and add value.
- ✓ Work on real life projects.
- ✓ Give a real life scenario and help to create applications more complicated and useful.
- ✓ Mentoring through email support.

We would like to send a great thank to our professor, Mr Le Thanh Nhan and others student for the adorable supports during the time in project.

Beside several mistakes we had made in the project, we hope to have more

opportunities to widen our knowledge in web developing world.

We also thank you so much for your information and the “demo” project from India APTECH to help us get more reality experience in this project. Since we have gained more experience from this project, we will not stop learning any thing as much as we can to be more successful in the future.

Your Sincerely,

Team Group 06.

Synopsis

The objective of this project is to develop a web-based Courier Management System named CourierXpress, which aims to automate courier operations such as booking, tracking, billing, and reporting.

This project provides students with hands-on experience in developing real-world applications using modern web technologies. Instead of focusing only on theory, the system allows students to implement practical solutions for real-life problems in the logistics industry.

The CourierXpress system is designed to support three main roles: Admin, Agent, and Customer. Each role has specific functionalities such as managing courier details, tracking shipments, handling billing processes, and generating reports.

Through this project, students will gain knowledge in database design, web development, system analysis, and application deployment. It also enhances problem-solving skills and prepares students for real-world IT environments.

Review 01

Design Plan: CourierXpress	Document Name: Problem Definition	SWD/Form No.01/PD/BVer1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 3 of 36

Problem Definition

In today's logistics industry, courier services face challenges in managing shipments, customer information, and delivery tracking efficiently. Many companies still rely on manual processes, which lead to errors, delays, and inefficiencies.

CourierXpress is designed to solve these problems by providing a centralized web-based system that automates courier operations.

The system will provide the following functionalities:

- Manage courier bookings with sender and receiver details
- Track shipment status in real-time
- Manage customer and agent information
- Generate billing and invoices automatically
- Provide reports and analytic for decision making
- Send notifications to customers about shipment updates

User Roles:

1. Admin:

- Manage users, agents, and couriers
- Generate reports and analytic
- Control system operations

2. Agent:

- Handle courier bookings
- Update delivery status
- Manage branch-specific data

3. Customer:

- Register and login
- Track courier using tracking ID
- View shipment history

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Design Plan: CourierXpress	Document Name: Customer Requirements Specifications	SWD/Form No.02/CRS/BVer1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 4 of 36

Customer Requirements Specifications

I. CRS/Requirements Specifications

Category	Description
Inputs	Customer registration data; branch and agent data; shipment booking data; status update data; billing information.
Outputs	Tracking details; shipment receipts and bills; shipment status updates; simple branch reports.
Processes	Authentication, customer management, shipment booking, shipment tracking, billing generation, and basic reporting.
Hardware / Software	Windows 10 or higher, Internet connection, browser, PHP/Laravel or equivalent back-end, MySQL 8.0 or higher.

S.No	Customer Acceptance Criteria
1	Given Admin, when managing couriers, then the system allows add, update, delete, assign, and bill-view operations.
2	Given Agent, when booking a shipment, then the system automatically generates a tracking number and shipment receipt.
3	Given Agent, when updating status, then the shipment timeline and current status are updated correctly.
4	Given Customer, when entering a valid tracking number, then the system displays shipment details and status history.

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Design Plan: CourierXpress	Document Name:Customer Requirements Specifications	SWD/Form No.02/CRS/BVer1.0
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II. Project Plan

- Client / Project Name: CourierXpress
- Project Duration: 08/04/2026 – 13/05/2026
- Vision / Objective: To design and develop a web-based courier management system that automates booking, tracking, billing, and basic operational management.
- Scope: Admin, Agent, and Customer modules; shipment management; billing; status tracking; and basic branch handling.
- Dependencies: Internet connection, VS Code, PHP/Laravel, MySQL, and modern web browsers.

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	Project Group No:06	LE THANH NHAN
Signature		
Date		

Review 02

Design Plan: CourierXpress	Document Name: Data Flow Diagram	SWD/Form No.03DFD/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 7 of 36

Data Flow Diagram

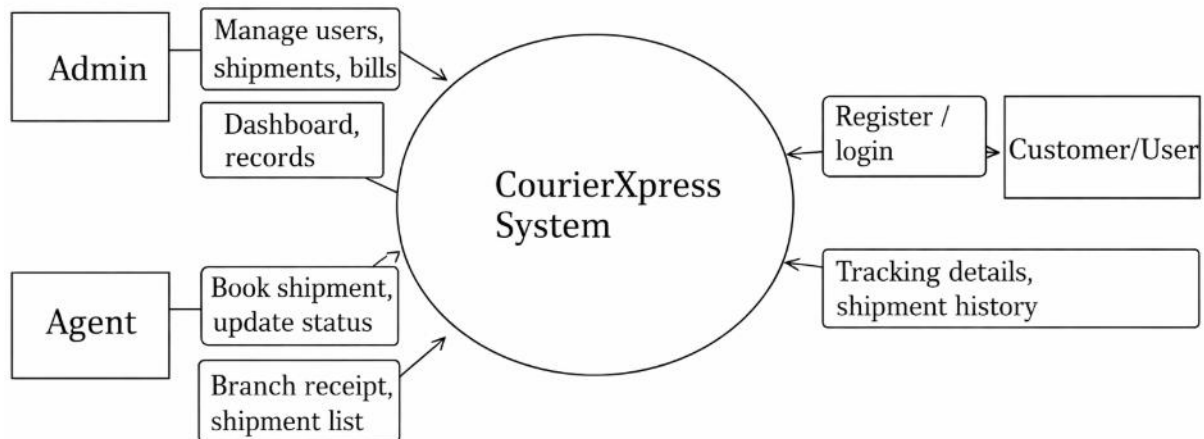


Figure 1. DFD Level 0

The Level 0 DFD presents the interaction between Admin, Agent, and Customer/User with the central CourierXpress system. It summarizes the high-level flows for management, booking, tracking, and billing.

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Design Plan: CourierXpress	Document Name: Data Flow Diagram	SWD/Form No.03DFD/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 8 of 36

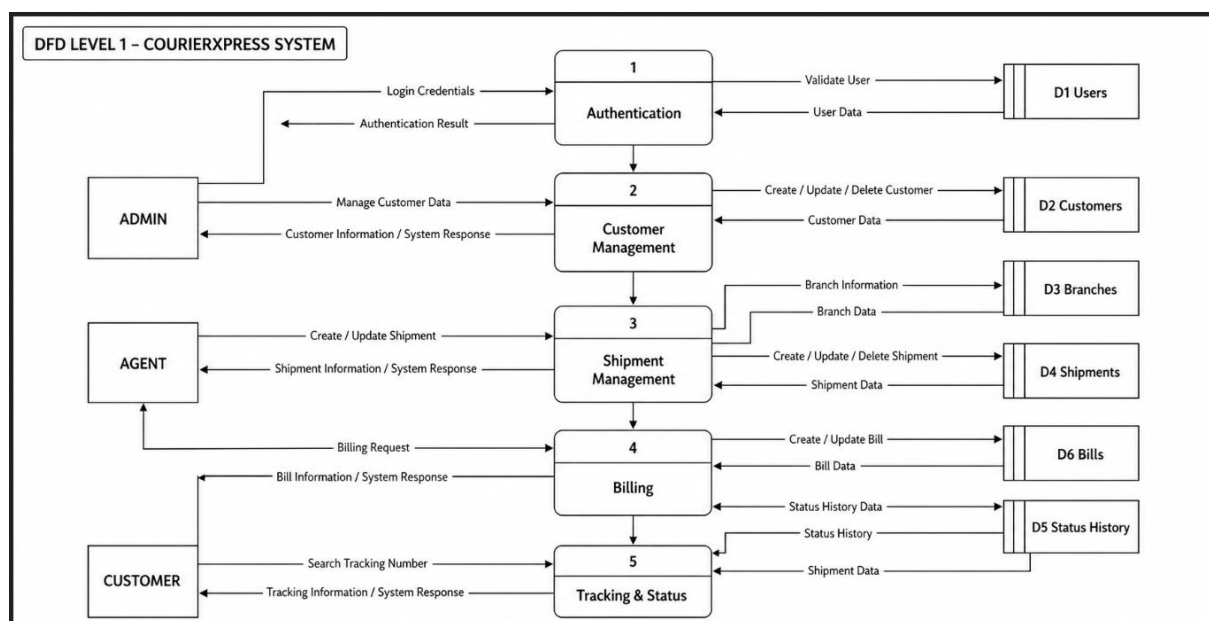


Figure 2. DFD Level 1

The Level 1 DFD decomposes the system into Authentication, Customer Management, Shipment Management, Billing, and Tracking & Status. These processes interact with Users, Customers, Branches, Shipments, Status History, and Bills data stores.

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Design Plan: CourierXpress	Document Name: Entity Relationship Diagram	SWD/Form No.04/ERD/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 9 of 36

A. Entity Relationship Diagram

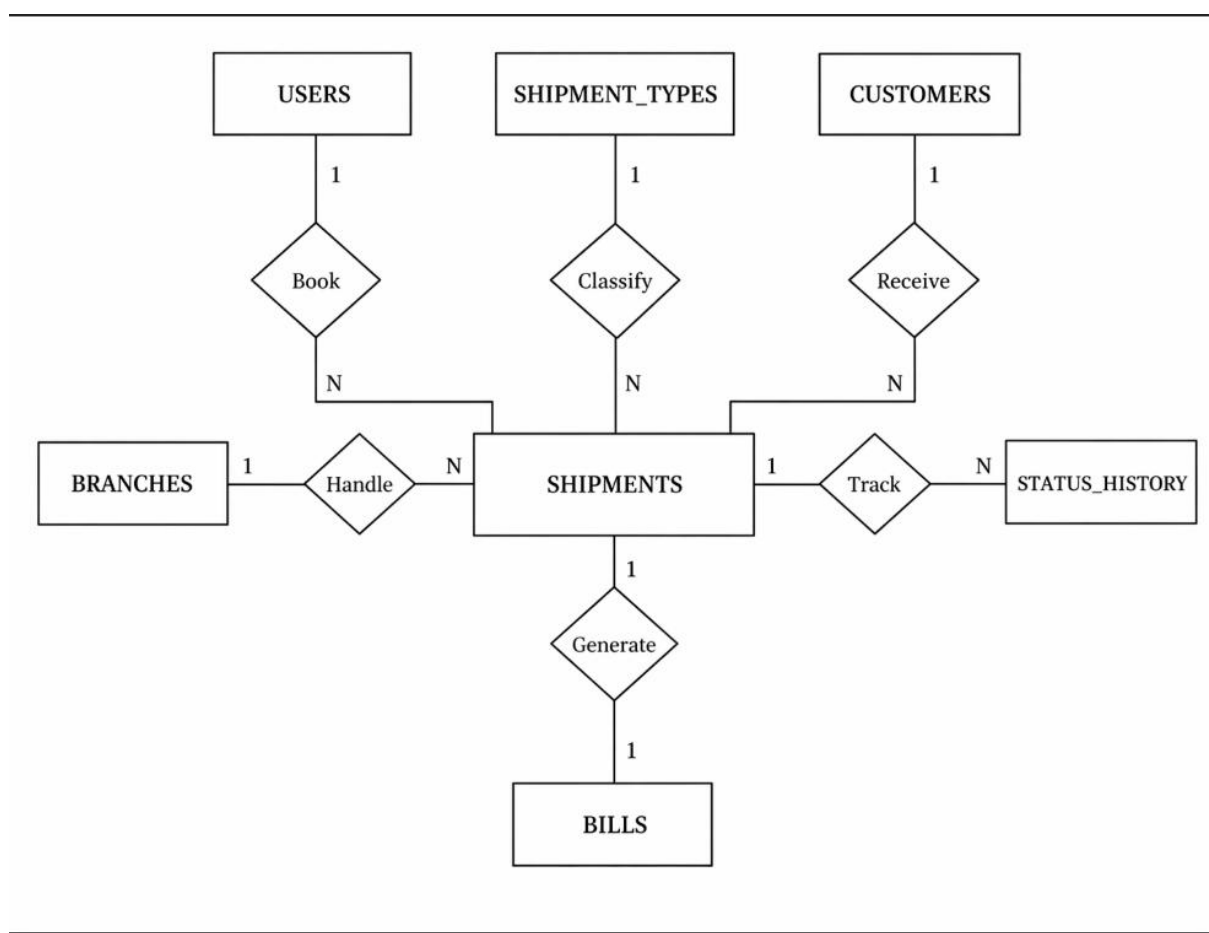


Figure 3. Conceptual ERD

The ERD identifies **SHIPMENTS** as the core entity. **USERS** book and manage shipments, **CUSTOMERS** send or receive shipments, **BRANCHES** handle operational processing, **SHIPMENT_TYPES** classify shipments, **STATUS_HISTORY** records shipment events, and **BILLS** are generated for each shipment.

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Design Plan: CourierXpress	Document Name: Entity Relationship Diagram	SWD/Form No.04/ERD/Ver1.0
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B. Entity Attributes

✓ Entity Properties:

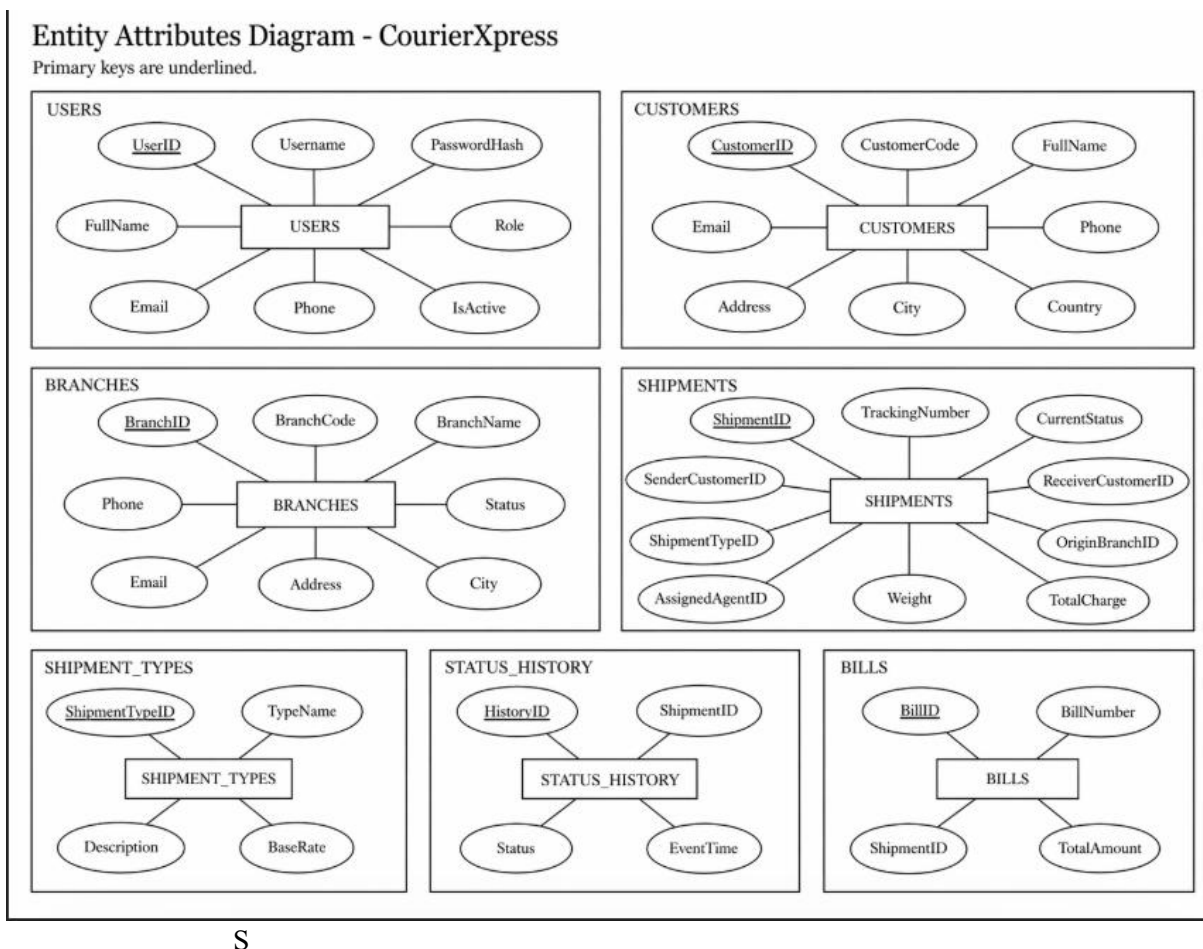


Figure 4. Entity Attributes

The entity-attribute diagrams list the key attributes for each major entity. Primary keys are underlined in the figure, and foreign keys are shown as relational attributes in the table design section.

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Design Plan: CourierXpress	Document Name: Table Design	SWD/Form No.05/TAB/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 11 of 36

A. Table Design

1.1.1 Table: branches

Field	Data Type	Key	Description
branch_id	BIGINT	PK	Branch identification
branch_code	VARCHAR(20)	Unique	Branch code
branch_name	VARCHAR(100)		Branch name
city	VARCHAR(100)		Branch city
status	VARCHAR(20)		Branch status

1.1.2 Table: users

Field	Data Type	Key	Description
user_id	BIGINT	PK	User identification
username	VARCHAR(50)	Unique	Login username
password_hash	VARCHAR(255)		Encrypted password
role	ENUM		ADMIN / AGENT / CUSTOMER
branch_id	BIGINT	FK	Assigned branch

1.1.3 Table: customers

Field	Data Type	Key	Description
customer_id	BIGINT	PK	Customer identification
user_id	BIGINT	FK	Linked account
customer_code	VARCHAR(20)	Unique	Customer code
full_name	VARCHAR(100)		Customer full name
phone	VARCHAR(20)		Contact number

1.1.4 Table: shipment_types

Field	Data Type	Key	Description
shipment_type_id	BIGINT	PK	Shipment type identification
type_name	VARCHAR(50)		Type name
base_rate	DECIMAL(12,2)		Base rate for pricing

1.1.5 Table: shipments

Field	Data Type	Key	Description
shipment_id	BIGINT	PK	Shipment identification
tracking_number	VARCHAR(50)	Unique	Tracking number
sender_customer_id	BIGINT	FK	Sender customer
receiver_customer_id	BIGINT	FK	Receiver customer
shipment_type_id	BIGINT	FK	Shipment type
origin_branch_id	BIGINT	FK	Origin branch

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Design Plan: CourierXpress	Document Name: Table Design	SWD/Form No.05/TAB/Ver1.0
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assigned_agent_id	BIGINT	FK	Assigned agent
current_status	ENUM		Current shipment status
total_charge	DECIMAL(12,2)		Total shipment charge

1.1.6 Table: shipment_status_history

Field	Data Type	Key	Description
history_id	BIGINT	PK	Status history identification
shipment_id	BIGINT	FK	Shipment reference
status	ENUM		Status name
event_time	DATETIME		Date/time of event

1.1.7 Table: bills

Field	Data Type	Key	Description
bill_id	BIGINT	PK	Bill identification
bill_number	VARCHAR(30)	Unique	Bill number
shipment_id	BIGINT	FK	Shipment reference
total_amount	DECIMAL(12,2)		Bill total amount

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Design Plan: CourierXpress	Document Name: Table Design	SWD/Form No.05/TAB/Ver1.0
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B. Relationship Diagram

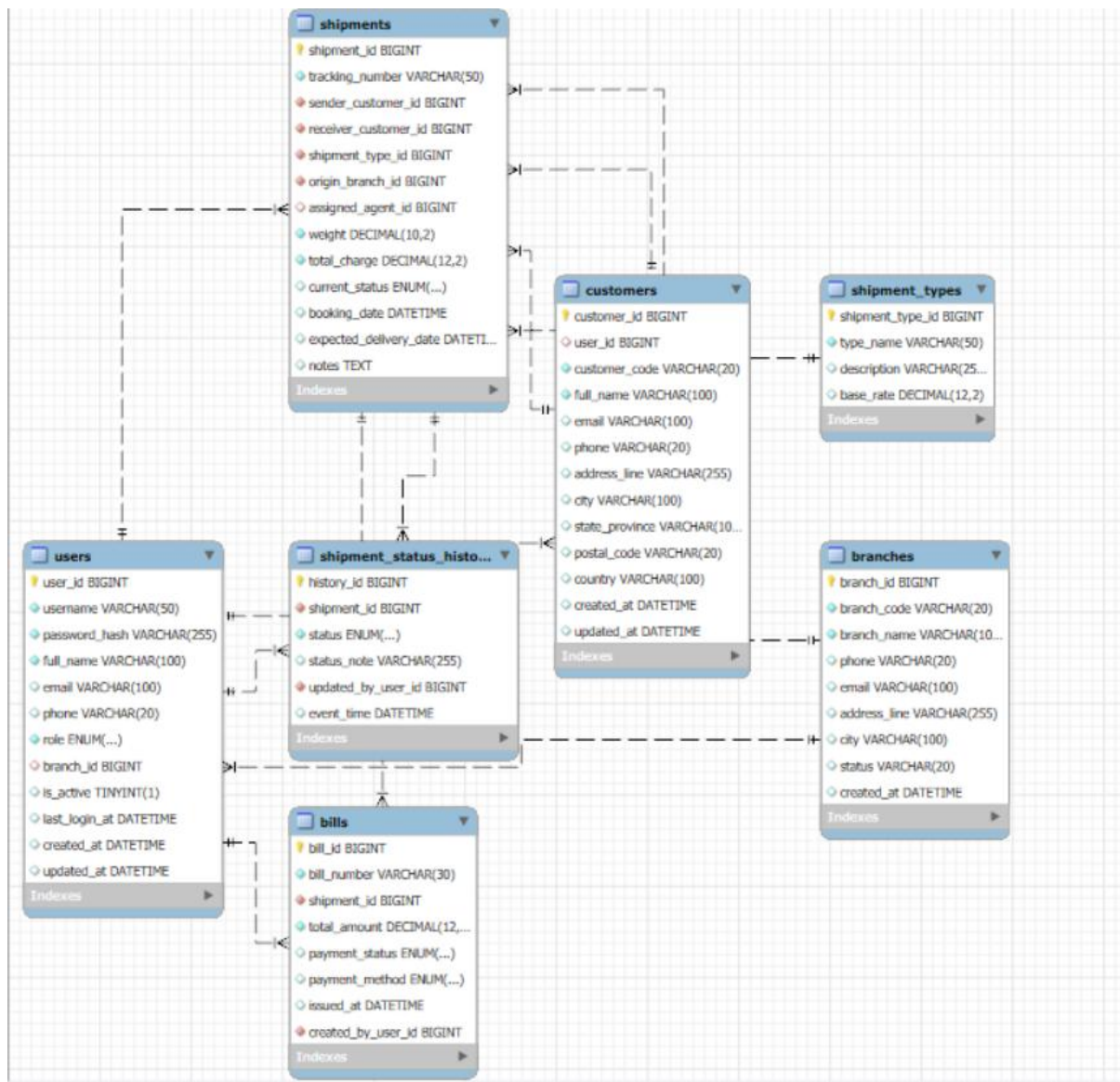


Figure 5. Physical Relationship Diagram

The relationship diagram shows the main physical tables and foreign-key links used in the MySQL schema. Users are linked to branches and customers, shipments link to customers, shipment types, branches, and assigned users, while shipment status history and bills depend on shipments.

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Design Plan: CourierXpress	Document Name: SiteMap	SWD/Form No.06/SM/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 14 of 36

SiteMap

Site Map

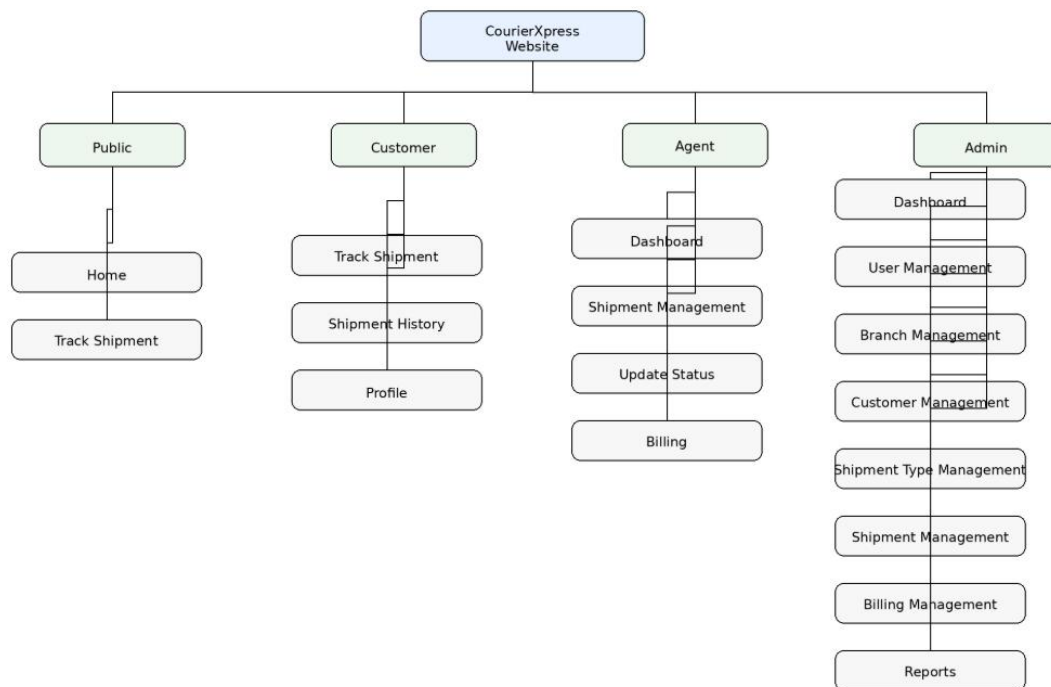


Figure 6. Site Map

The site map defines the navigation structure of the web application. From the landing page, users can reach login, registration, tracking, the agent portal, and the admin portal. Each role-specific dashboard provides access to its own operational pages.

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Design Plan: CourierXpress	Document Name: Algorithm	SWD/Form No.07/ALG/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 15 of 36

Algorithm

Algorithm - Admin

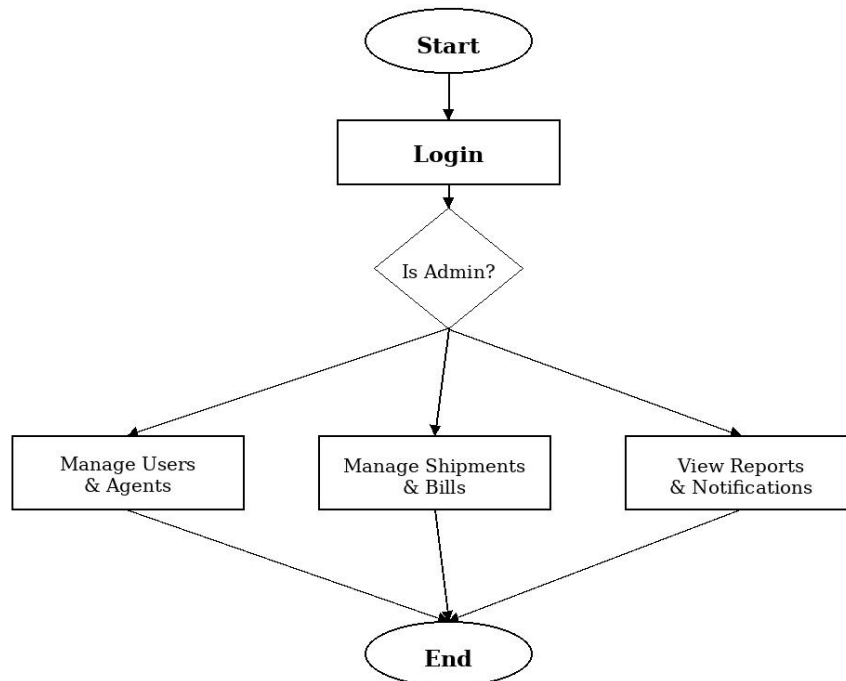


Figure 7. Admin Algorithm

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Date		

Design Plan: CourierXpress	Document Name: Algorithm	SWD/Form No.07/ALG/Ver1.0
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Algorithm - Agent

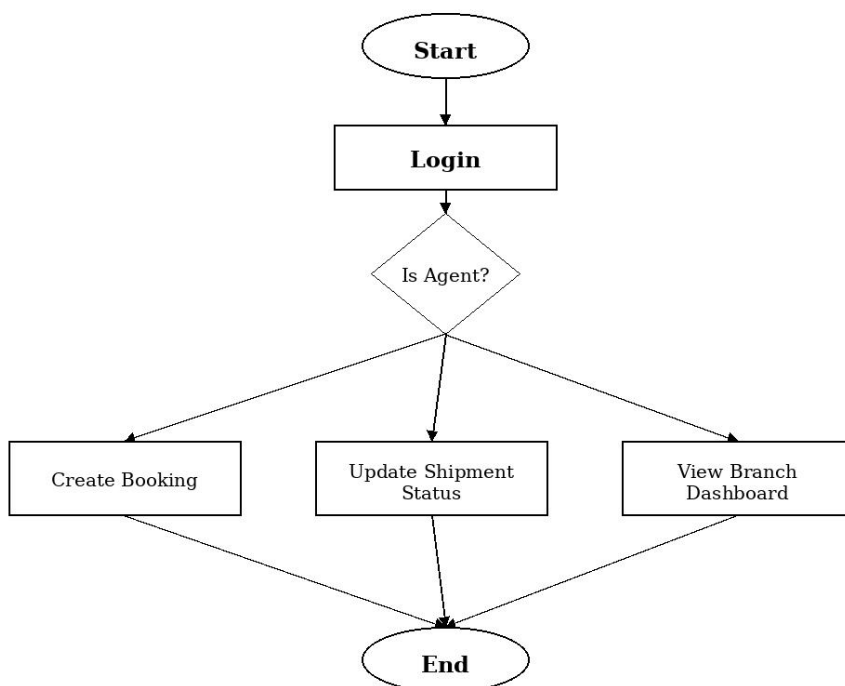


Figure 8. Agent Algorithm

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Signature		
Date		

Design Plan: CourierXpress	Document Name: Algorithm	SWD/Form No.07/ALG/Ver1.0
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Algorithm - Customer

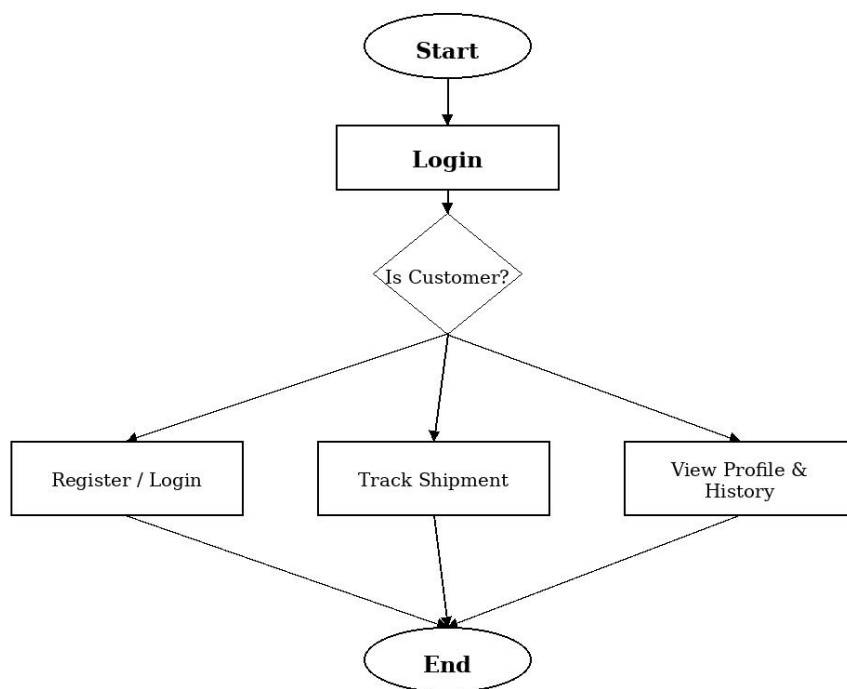


Figure 9. Customer Algorithm

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Date		

Review 03

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 19 of 36

Screen shots

1.1.8 Login Page

Login Page

Login Page

CourierXpress

Username or Email

Password

[Forgot password?](#)

[Register](#)

Description: This screen allows Admin, Agent, and Customer users to sign in using account credentials.

From: Home / Landing Page

To: Admin Dashboard, Agent Dashboard, Customer Dashboard

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Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
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1.1.9 Admin Dashboard

Admin Dashboard

Admin Dashboard

Sidebar
Dashboard
Shipments
Customers
Agents
Bills
Reports

Total Shipments

In Transit

Delivered

Delayed

Tracking No	Sender	Receiver	Status	Branch

New Shipment

Description: This layout summarizes system statistics and provides quick access to shipment management, customers, agents, bills, and reports.

From: Login Page

To: Shipment Management, Reports, User Management

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Signature		
Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
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1.1.10 Track Shipment Page

Track Shipment Page

Track Shipment Page

Enter Tracking Number

Track

Shipment Summary

Tracking No | Current Status | Expected Delivery

Date/Time	Location	Status	Remark

Quick Info

Sender
Receiver
Type
Weight
Charge

Description: This page allows customers to enter a tracking number and review shipment summary and timeline information.

From: Home / Landing Page or Customer Dashboard

To: Shipment History

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	Project Group No:06	LE THANH NHAN
Signature		
Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 22 of 36

1.1.11 Customer Management Page

Customer Management Page

Dashboard

Customers

Shipments

Bills

Reports

Search customer by name / phone / city

Code	Name	Phone	City	Action

Description: This screen allows Admin users to manage customer records, search customers by name, phone, or city, and perform add, edit, or delete operations.

From: Admin Dashboard

To: Add Customer Page, Edit Customer Page, Shipment Management Page

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Signature		
Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 23 of 36

1.1.12 Shipment Management Page

Shipment Management Page

Dashboard

Bookings

Status

Bills

Tracking

Filter by tracking no / status / branch

Tracking	Sender	Receiver	Status	Branch	Action

Description: This screen allows Admin or Agent users to manage shipments, filter shipments by tracking number, status, or branch, and access shipment details.

From: Admin Dashboard / Agent Dashboard

To: Create Shipment Booking Page, Update Shipment Status Page, Billing Page

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Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
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1.1.13 Create Shipment Booking Page

Create Shipment Booking Page

Sender Information

Full Name

Phone

Address

Receiver Information

Full Name

Phone

Address

Shipment Details

Type

Weight

Branch

Expected Date

Tracking Number: auto-generated

Create

Reset

Description: This screen allows Agent users to create a new shipment booking by entering sender information, receiver information, and shipment details. The system automatically generates a tracking number after submission.

From: Shipment Management Page / Agent Dashboard

To: Shipment Receipt Page, Shipment Management Page

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Signature		
Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
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1.1.14 Update Shipment Status Page

Update Shipment Status Page

Enter tracking number to update status

Shipment Summary

Tracking No	Sender	Current Status	Branch

Update Status

New Status

Remark

Status Timeline

Date/Time	Location	Status	Remark

Description: This screen allows Agent users to search for a shipment by tracking number, update the current shipment status, and record remarks in the status timeline.

From: Shipment Management Page / Agent Dashboard

To: Shipment History Page, Shipment Management Page

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Signature		
Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
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1.1.15 Billing / Bill Details Page

Billing / Bill Details Page

Bill Search

Bill No / Tracking No

Bill Information

Bill No	
Tracking No	
Customer	
Total Amount	
Payment Status	
Payment Method	

Actions

Bill Items / Summary

Service	Qty	Unit Price	Amount

Description: This screen allows Admin or Agent users to search bills, view bill details, update payment status, and print or export a bill.

From: Admin Dashboard / Agent Dashboard / Shipment Management Page

To: Bill Print Preview, Shipment Management Page

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Signature		
Date		

DesignPlan: CourierXpress	Document Name: Screen shots	SWD/Form No.08/SS/Ver1.0
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1.1.16 Customer Profile & Shipment History

Customer Profile & Shipment History

Profile Information

Full Name

Phone

Email

Address

Update

Shipment History

Tracking	Date	Status	Expected	Action

View

Description: This screen allows Customer users to update profile information and review the history of their previous shipments.

From: Customer Dashboard / Track Shipment Page

To: Track Shipment Page, Profile Update Confirmation

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Signature		
Date		

DesignPlan: CourierXpress	Document Name: Testing document	SWD/Form No.09/TD/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 28 of 36

Testing document

Sr.No	Features Tested	Remarks
1	Login with valid credentials	Pass
2	Login with invalid credentials	Pass
3	Register a new customer account	Pass
4	Create new shipment booking	Pass
5	Auto-generate tracking number	Pass
6	Update shipment status	Pass
7	Display shipment status history	Pass
8	Generate bill for shipment	Pass
9	View branch dashboard statistics	Pass
10	Search shipments by tracking number	Pass
11	Track shipment from public page	Pass
12	View bill details for shipment	Pass
13	Filter customers by phone or city	Pass
14	Filter shipments by current status	Pass
15	Filter shipments by branch	Pass

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Design Plan: CourierXpress	Document Name: Final checklist	SWD/Form No. 10/FC/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 29 of 36

Final checklist

Sr.No	Aspect Tested	Suggestion / Remarks
1	Are all the users able to view the images and links?	OK / Ready for submission
2	Have all the views, modules and controllers been properly integrated?	OK / Ready for submission
3	Are the GUI contents devoid of spelling mistakes?	OK / Ready for submission
4	Is the application user-friendly?	OK / Ready for submission
5	Is the website launching correctly in tested browsers?	OK / Ready for submission
6	Are all the forms validated with proper criteria?	OK / Ready for submission
7	Do all text links lead to the appropriate page?	OK / Ready for submission
8	Do all image links lead to the appropriate page?	OK / Ready for submission
9	Are all the images and links clearly visible on the page?	OK / Ready for submission
10	Does the web page work properly in all the tested browsers?	OK / Ready for submission
11	Does the web page take too long to be loaded fully?	OK / Ready for submission
12	Is the navigation sequence correct through all web pages?	OK / Ready

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Design Plan: CourierXpress	Document Name: Final checklist	SWD/Form No.10/FC/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 30 of 36

		for submission
13	Is the JavaScript code working as expected in click events?	OK / Ready for submission
Sr.No	Aspect Tested	Suggestion / Remarks
1	Are all the users able to view the images and links?	OK / Ready for submission

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Date		

Design Plan:	Document Name: Task Sheet	SWD/Form No.11/TS/Ver1.0
Effective Date: 10/4/2026	Version 1.0	Page No: 31 of 36

No	Customer	Role	Contents	Table Name	Planed Start Date	Actual Start Date	Actual Days	Status
1	Phan Văn Duy	Admin	Branch Management	Branches	08/04/2026	08/04/2026	5	OK
2		Admin	Customer Management	Customers	08/04/2026	08/04/2026	5	OK
3		User	Register, login, Forgot password	Users, Customers	10/04/2026	10/04/2026	5	OK
4		User	Personal Account Management	Users, Customers	10/04/2026	10/04/2026	5	OK
1	Lê Hoàng Thông	Agent	Shipment Management, create shipment	Shipments	15/04/2026	15/04/2026	5	OK
2		Agent	Update Shipment Status	Shipments, Shipment Status History	15/04/2026	15/04/2026	5	OK
3		User	Track Shipment	Shipments	17/04/2026	17/04/2026	5	OK
4		User	Shipment History	Shipment Status History	17/04/2026	17/04/2026	5	OK
1	Vương Quế Hân	Admin	User Management	User	N/A	N/A	N/A	N/A
2		Admin	Shipment Type Management	Shipment_Types	N/A	N/A	N/A	N/A
3		Admin	Billing Management	Bills	N/A	N/A	N/A	N/A
4		User	View / Print Bill	Bills	N/A	N/A	N/A	N/A

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	Project Group No:06	LE THANH NHAN
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Date		